Class 9 Mathematics — Exercise 5.2

Basic Concept

(1) Radical Equation

An equation in which the unknown (variable) appears **under a radical sign** is called a radical equation.

Example: $\sqrt{x+1} = 7$

(2) Linear Equation

An equation in which the exponent/power of the unknown (variable) is **always 1** is called a linear equation.

Example: $x + 1 = 7$

(3) Extraneous Root

While squaring both sides of a radical equation, it is possible to get an extra root that **does not satisfy the original equation**. Such a root is called an **extraneous root**. So it is necessary to check every root by putting it into the original equation.

Example: $4 + 2\sqrt{3y+1} = 3$

$$2\sqrt{3y+1} = -1$$

$$\left(2\sqrt{3y+1}\right)^2 = (-1)^2$$

$$4(3y+1) = 1$$

$$12y + 4 = 1$$

$$12y = -3 \quad \Rightarrow \quad y = -\frac{1}{4}$$

Check:

$$4 + 2\sqrt{3\left(-\tfrac{1}{4}\right) + 1} \stackrel{?}{=} 3$$

$$4 + 2\sqrt{-\tfrac{3}{4} + 1} \stackrel{?}{=} 3$$

$$4 + 2\sqrt{\tfrac{1}{4}} \stackrel{?}{=} 3$$

$$4 + 1 \stackrel{?}{=} 3$$

$$5 \neq 3$$

Thus $y = -\frac{1}{4}$ is an extraneous root and the solution set is ϕ.

Key Fact

When the **radical term is equal to a negative number**, such an equation has **no solution in real numbers**: $S.S = \{\ \}$ or ϕ.

Useful Points

$$\sqrt{\ } = (\ )^{\frac{1}{2}} \qquad \left(\sqrt{a}\right)^2 = a \qquad 4^2 = 4 \times 4 \qquad (-) \times (-) = (+)$$

Method used in every question: Isolate the radical term → Taking square on both sides (b/s) → Solve the linear equation → Check → S.S

Question 1

$$\sqrt{2x} = 4$$

Solution: Taking square on b/s

$$\left(\sqrt{2x}\right)^2 = (4)^2$$
$$2x = 16$$
$$x = \frac{16}{2}$$
$$\boxed{x = 8}$$

Check:

$$\sqrt{2x} = 4$$
$$\sqrt{2(8)} = 4$$
$$\sqrt{16} = 4$$
$$4 = 4$$

$$\boxed{S.S = \{8\}}$$

Question 2

$$\sqrt{x-3} = 2$$

Solution: Taking square on b/s

$$\left(\sqrt{x-3}\right)^2 = (2)^2$$
$$x - 3 = 4$$
$$x = 4 + 3$$
$$\boxed{x = 7}$$

Check:

$$\sqrt{x-3} = 2$$
$$\sqrt{7-3} = 2$$

$$\sqrt{4} = 2$$
$$2 = 2$$

$$\boxed{S.S = \{7\}}$$

Question 3

$$\sqrt{x-5} = 3$$

Solution: Taking square on b/s

$$\left(\sqrt{x-5}\right)^2 = (3)^2$$
$$x - 5 = 9$$
$$x = 9 + 5$$
$$\boxed{x = 14}$$

Check:

$$\sqrt{x-5} = 3$$
$$\sqrt{14-5} = 3$$
$$\sqrt{9} = 3$$
$$3 = 3$$

$$\boxed{S.S = \{14\}}$$

Question 4

$$\sqrt{2x+1} = 9$$

Solution: Taking square on b/s

$$\left(\sqrt{2x+1}\right)^2 = (9)^2$$
$$2x + 1 = 81$$
$$2x = 81 - 1$$
$$2x = 80$$
$$x = \frac{80}{2}$$
$$\boxed{x = 40}$$

Check:

$$\sqrt{2x+1} = 9$$
$$\sqrt{2(40)+1} = 9$$
$$\sqrt{80+1} = 9$$

$$\sqrt{81} = 9$$
$$9 = 9$$

$$\boxed{S.S = \{40\}}$$

Question 5

$$\sqrt{5x-4} = 14$$

Solution: Taking square on b/s

$$\left(\sqrt{5x-4}\right)^2 = (14)^2$$
$$5x - 4 = 196$$
$$5x = 196 + 4$$
$$5x = 200$$
$$x = \frac{200}{5}$$
$$\boxed{x = 40}$$

Check:

$$\sqrt{5x-4} = 14$$
$$\sqrt{5(40)-4} = 14$$
$$\sqrt{200-4} = 14$$
$$\sqrt{196} = 14$$
$$14 = 14$$

$$\boxed{S.S = \{40\}}$$

Question 6

$$\sqrt{3x-5} = -10$$

When the radical term is equal to a negative number, such an equation has no solution in real numbers.

$$S.S = \{\ \} \text{ or } \phi$$

Solving it to see why: Taking square on b/s

$$\left(\sqrt{3x-5}\right)^2 = (-10)^2 \qquad (\because (-) \times (-) = +)$$
$$3x - 5 = 100$$
$$3x = 100 + 5$$
$$3x = 105$$

$$x = \frac{105}{3}$$

$$\boxed{x = 35}$$

Check:

$$\sqrt{3x-5} = -10$$
$$\sqrt{3(35)-5} \stackrel{?}{=} -10$$
$$\sqrt{105-5} \stackrel{?}{=} -10$$
$$\sqrt{100} \stackrel{?}{=} -10$$
$$10 \neq -10$$

$x = 35$ is an extraneous root.

$$\boxed{S.S = \{\ \} \text{ or } \phi}$$

Question 7

$$\sqrt{y+4} - 3 = 2$$

Solution:

$$\sqrt{y+4} = 2 + 3$$
$$\sqrt{y+4} = 5$$

Taking square on b/s

$$\left(\sqrt{y+4}\right)^2 = (5)^2$$
$$y + 4 = 25$$
$$y = 25 - 4$$
$$\boxed{y = 21}$$

Check:

$$\sqrt{y+4} - 3 = 2$$
$$\sqrt{21+4} - 3 = 2$$
$$\sqrt{25} - 3 = 2$$
$$5 - 3 = 2$$
$$2 = 2$$

$$\boxed{S.S = \{21\}}$$

Question 8

$$5 - \sqrt{2x-1} = 0$$

Solution:

$$5 = \sqrt{2x-1}$$
$$\Rightarrow \sqrt{2x-1} = 5$$

Taking square on b/s

$$\left(\sqrt{2x-1}\right)^2 = (5)^2$$
$$2x - 1 = 25$$
$$2x = 25 + 1$$
$$2x = 26$$
$$x = \frac{26}{2}$$
$$\boxed{x = 13}$$

Check:

$$5 - \sqrt{2x-1} = 0$$
$$5 - \sqrt{2(13)-1} = 0$$
$$5 - \sqrt{26-1} = 0$$
$$5 - \sqrt{25} = 0$$
$$5 - 5 = 0$$
$$0 = 0$$

$$\boxed{S.S = \{13\}}$$

Question 9

$$\sqrt{y+1} - 12 = -10$$

Solution:

$$\sqrt{y+1} = -10 + 12$$
$$\sqrt{y+1} = 2$$

Taking square on b/s

$$\left(\sqrt{y+1}\right)^2 = (2)^2$$
$$y + 1 = 4$$
$$y = 4 - 1$$

$$\boxed{y = 3}$$

Check:

$$\sqrt{y+1} - 12 = -10$$
$$\sqrt{3+1} - 12 = -10$$
$$\sqrt{4} - 12 = -10$$
$$2 - 12 = -10$$
$$-10 = -10$$

$$\boxed{S.S = \{3\}}$$

Question 10

$$\sqrt{5t-2} = \sqrt{3t+4}$$

Solution: Taking square on b/s

$$\left(\sqrt{5t-2}\right)^2 = \left(\sqrt{3t+4}\right)^2$$
$$5t - 2 = 3t + 4$$
$$5t - 3t = 4 + 2$$
$$2t = 6$$
$$t = \frac{6}{2}$$
$$\boxed{t = 3}$$

Check:

$$\sqrt{5t-2} = \sqrt{3t+4}$$
$$\sqrt{5(3)-2} = \sqrt{3(3)+4}$$
$$\sqrt{15-2} = \sqrt{9+4}$$
$$\sqrt{13} = \sqrt{13}$$

$$\boxed{S.S = \{3\}}$$

Question 11

$$\sqrt{9-2x} = \sqrt{5x-12}$$

Solution: Taking square on b/s

$$\left(\sqrt{9-2x}\right)^2 = \left(\sqrt{5x-12}\right)^2$$
$$9-2x = 5x-12$$
$$9+12 = 5x+2x$$
$$21 = 7x$$
$$\frac{21}{7} = x$$
$$3 = x$$
$$\Rightarrow \boxed{x=3}$$

Check:

$$\sqrt{9-2x} = \sqrt{5x-12}$$
$$\sqrt{9-2(3)} = \sqrt{5(3)-12}$$
$$\sqrt{9-6} = \sqrt{15-12}$$
$$\sqrt{3} = \sqrt{3}$$

$$\boxed{S.S = \{3\}}$$

Question 12

$$12 - \sqrt{y+1} = 14$$

Solution:

$$-\sqrt{y+1} = 14-12$$
$$-\sqrt{y+1} = 2$$
$$\sqrt{y+1} = -2$$

When the radical term is equal to a negative number, then such an equation has no solution in real numbers.

Taking square on b/s

$$\left(\sqrt{y+1}\right)^2 = (-2)^2$$
$$y+1 = 4$$
$$y = 4-1$$
$$\boxed{y=3}$$

Check:

$$12 - \sqrt{y+1} = 14$$
$$12 - \sqrt{3+1} \stackrel{?}{=} 14$$
$$12 - \sqrt{4} \stackrel{?}{=} 14$$
$$12 - 2 \stackrel{?}{=} 14$$
$$10 \neq 14$$

So $y = 3$ is an extraneous root.

$$\boxed{S.S = \{\ \} \text{ or } \phi}$$

Question 13

$$4\sqrt{z} + 8 = 40$$

Solution:

$$4\sqrt{z} = 40 - 8$$
$$4\sqrt{z} = 32$$
$$\sqrt{z} = \frac{32}{4}$$
$$\sqrt{z} = 8$$

Taking square on b/s

$$\left(\sqrt{z}\right)^2 = (8)^2$$
$$\boxed{z = 64}$$

Check:

$$4\sqrt{z} + 8 = 40$$
$$4\sqrt{64} + 8 = 40$$
$$4(8) + 8 = 40$$
$$32 + 8 = 40$$
$$40 = 40$$

$$\boxed{S.S = \{64\}}$$

Question 14

$$\sqrt{\frac{a+6}{a+2}} = \sqrt{\frac{a+2}{a-1}}$$

Solution: Taking square on b/s

$$\left(\sqrt{\frac{a+6}{a+2}}\right)^2 = \left(\sqrt{\frac{a+2}{a-1}}\right)^2$$
$$\frac{a+6}{a+2} = \frac{a+2}{a-1}$$

Cross multiplication:

$$(a+6)(a-1) = (a+2)(a+2)$$
$$a(a-1)+6(a-1) = a(a+2)+2(a+2)$$
$$a^2 - a + 6a - 6 = a^2 + 2a + 2a + 4$$
$$a^2 + 5a - 6 = a^2 + 4a + 4$$
$$a^2 + 5a - a^2 - 4a = 4 + 6$$
$$\boxed{a = 10}$$

Check:

$$\sqrt{\frac{a+6}{a+2}} = \sqrt{\frac{a+2}{a-1}}$$
$$\sqrt{\frac{10+6}{10+2}} = \sqrt{\frac{10+2}{10-1}}$$
$$\sqrt{\frac{16}{12}} = \sqrt{\frac{12}{9}}$$
$$\sqrt{\frac{4}{3}} = \sqrt{\frac{4}{3}}$$

$$\boxed{S.S = \{10\}}$$

Question 15

$$\sqrt{\frac{z}{z+3}} = \sqrt{\frac{z+2}{z+6}}$$

Solution: Taking square on b/s

$$\left(\sqrt{\frac{z}{z+3}}\right)^2 = \left(\sqrt{\frac{z+2}{z+6}}\right)^2$$
$$\frac{z}{z+3} = \frac{z+2}{z+6}$$

Cross multiplication:

$$z(z+6) = (z+2)(z+3)$$
$$z^2 + 6z = z(z+3) + 2(z+3)$$
$$z^2 + 6z = z^2 + 3z + 2z + 6$$
$$z^2 + 6z = z^2 + 5z + 6$$
$$z^2 + 6z - z^2 - 5z = 6$$
$$\boxed{z = 6}$$

Check:

$$\sqrt{\frac{z}{z+3}} = \sqrt{\frac{z+2}{z+6}}$$
$$\sqrt{\frac{6}{6+3}} = \sqrt{\frac{6+2}{6+6}}$$
$$\sqrt{\frac{6}{9}} = \sqrt{\frac{8}{12}}$$
$$\sqrt{\frac{2}{3}} = \sqrt{\frac{2}{3}}$$

$$\boxed{S.S = \{6\}}$$

Question 16

$$\sqrt{5x-4} = \sqrt{7x+2}$$

Solution: Taking square on b/s

$$\left(\sqrt{5x-4}\right)^2 = \left(\sqrt{7x+2}\right)^2$$
$$5x - 4 = 7x + 2$$
$$5x - 7x = 2 + 4$$
$$-2x = 6$$
$$x = \frac{6}{-2}$$
$$\boxed{x = -3}$$

Check:

$$\sqrt{5x-4} = \sqrt{7x+2}$$
$$\sqrt{5(-3)-4} = \sqrt{7(-3)+2}$$
$$\sqrt{-15-4} = \sqrt{-21+2}$$
$$\sqrt{-19} = \sqrt{-19}$$

$$\boxed{S.S = \{-3\}}$$